

Col-AT-1
Pseudocode writing.

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1. Design Complete + a class hierarchy.

START

CLASS Person

VARIABLE name

VARIABLE age

METHOD getPersonDetails()

INPUT name

INPUT age

END METHOD

END CLASS

CLASS Student INHERITS Person

VARIABLE regNo

ARRAY marks[5]

METHOD getStudentDetails()

INPUT regNo

FOR $i = 1$ TO 5

INPUT marks[i]

END FOR

END METHOD

METHOD getStudentDetails()

INPUT regNo

SUM $\leftarrow 0$

FOR $i = 1$ TO 5

SUM $\leftarrow$ SUM + marks[i]

END FOR

average $\leftarrow$ SUM / 5

RETURN average
END METHOD

END CLASS

CLASS ResearchStudent INHERITS Student

VARIABLE ?researchArea

METHOD getResearchDetails()

INPUT researchArea

END METHOD

METHOD displayCompleteDetails()

PRINT "Name: ", name

PRINT "Age: ", age

PRINT "Register Number: ", regNo

PRINT "Marks: ", marks

PRINT "Average: ",

CalculateAverage()

PRINT "Research Area: ",

?researchArea

END METHOD

END CLASS

MAIN

CREATE OBJECT rs OF ResearchStudent

CALL rs.getPersonDetails()

CALL rs.getStudentDetails()

CALL rs.getResearchDetails()

CALL rs.CalculateAverage()

CALL rs.displayCompleteDetails()

END

2.

```
BEGIN
READ N
READ ARRAY A
FOR i ← 0 TO N-1
  FOR j ← i+1 TO N-1
    IF A[i] == A[j]
      DELETE A[j]
    ENDIF
  ENDFOR
ENDFOR
PRINT Modified Array
END.
```

3. Debug the Pseudocode

BEGIN

CLASS Area

METHOD area(side)

RETURN side * side

END METHOD

METHOD area(length, breadth)

RETURN length * breadth

END METHOD

END CLASS

END

4. Trace the Execution.

BEGIN

Sum $\leftarrow$ 0

For $i \leftarrow 1$ TO 5

IF $i \bmod 2 = 0$

Sum $\leftarrow$ Sum + i

ELSE

Sum $\leftarrow$ Sum - i

END IF

PRINT Sum

END

5. Write Efficient Pseudocode:-

BEGIN

READ N

READ Array A

First $\leftarrow -\infty$

Second $\leftarrow -\infty$

FOR $i \leftarrow 0$ TO $N-1$

IF $A[i] > \text{First}$ THEN

Second $\leftarrow \text{First}$

First $\leftarrow A[i]$

ELSE IF $A[i] > \text{Second}$ AND $A[i] \neq$

First THEN

Second $\leftarrow A[i]$

END IF

END FOR

PRINT Second

END

6. Pseudocode

```
BEGIN
INTERFACE Payment
    METHOD Pay(amount)
END INTERFACE

CLASS Creditcard IMPLEMENTS Payment
    METHOD Pay (amount)
        PRINT "Paid using credit card"
    END METHOD
END CLASS

CLASS UPI IMPLEMENTS Payment
    METHOD Pay (amount)
        PRINT "Paid using UPI"
    END METHOD
END CLASS

CLASS NetBanking UPI IMPLEMENTS Payment
    METHOD Pay (amount)
        PRINT "Paid using Net Banking"
    END METHOD
END CLASS

MAIN
    READ choice, amount

    IF choice = 1
        P = credit-card
    ELSE IF choice = 2
        P = UPI
    ELSE
        P = NetBanking
    ENDIF
    CALL P.Pay (amount)
END
```

7. CLASS Animal

METHOD Speak()

PRINT "Animal"

END METHOD

END CLASS

CLASS Dog INHERITS Animal

METHOD Speak()

PRINT "Barks"

END METHOD

END CLASS

Animal obj ← NEW Dog()

Obj.Speak()

END

8.

BEGIN

CLASS Employee

DATA

id

name

CONSTRUCTOR(id, name)

this.id ← id

this.name ← name

END CONSTRUCTOR

METHOD Display()

PRINT "Employee ID: ", id

PRINT "Employee Name: ", Name

END METHOD

END CLASS

Employee e ← New Employee(101, "John")

CALL e.Display()

END.

9. BEGIN

CLASS Vehicle

DATA

VehicleNo

ownerName

METHOD display()

PRINT VehicleNo

PRINT ownerName

END METHOD

END CLASS

CLASS Car INHERITS Vehicle

DATA

-fuelType

METHOD display()

PRINT VehicleNo

PRINT ownerName

PRINT -fuelType

END METHOD

END CLASS

CLASS Truck INHERITS Vehicle

DATA

loadCapacity

METHOD display()

PRINT VehicleNo

PRINT ownerName

PRINT loadCapacity

END METHOD

END CLASS

MAIN

READ N

FOR $i \leftarrow 1$ TO N

 READ type

 IF type = "Car"

 CREATE Car C

 READ VehicleNo, OwnerName, fuelType

 STORE C

 ELSE

 CREATE Truck T

 READ VehicleNo, OwnerName,

 load Capacity

 STORE T

 ENDIF

ENDFOR

FOR EACH Vehicle V

 CALL V.display()

ENDFOR

END